

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

LOCAL MANAGEMENT INSTRUCTION
LMI/OGN/ELEC/005
ISSUE NO.: 1, REVISION-5, DATE: Oct 2023

GENERATOR ROTOR END WINDING
RETAINING RINGS

APPROVED FOR

IMPLEMENTATION by _____
GROUP GENERAL MANAGER (SMTTP)

DATE _____

NTPC LTD.

SIMHADRI SUPER THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	GENERATOR ROTOR END WINDING RETAINING RINGS		
LMI No.	LMI/OGN/ELEC/005		
ISSUE NO.	1	Revision : 5	Date :Oct 2023

	DESIGNATION	SIGNATURE & NAME
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Checked by	AGM (EM)	
Reviewed by	AGM (EM) I/C	
Reviewed by	HOD(Chemistry)	
Reviewed by	GM (Operation)	
Reviewed by	GM (Maint)	
Reviewed by	GM(O&M)	

GENERATOR ROTOR END WINDING RETAINING RINGS

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1.0**OBJECTIVE:**

Failures of Generator Rotor Retaining Rings have been experienced on number of occasions. Objective is to take various measures to avoid the failure due to:

- a) Due to combination of stress in components and exposure to an aggressive environment, including pure water.
- b) Due to fretting which is a wear/corrosion mechanism resulting from the slight relative movement of contacting surfaces.
- c) Cracking

To safeguard against these, precautions during Operation & Maintenance is required to be taken.

RINGS IN SERVICE:

Relative humidity inside Generator has to be controlled. Inside Generator casing the cooling media dew point is to be controlled under all condition of load or no load. This can be done by maintaining H2 Gas dew point temperature below 0 deg c at frame pressure. Cold gas temperature is to be maintained above 20°C. (Cold gas temperature being higher).

RINGS DISMANTLED:

Removed rings should be kept in containers, after wrapping the ring with terylene reinforced polythene wrapped around. A desiccant for moisture absorption in sealed bag to be placed inside container. Any scratch mark or mechanical damaged should be avoided.

2.0**SCOPE:**

The Document covers the Rotor Retaining Rings for Generators of Unit – 1, 2, 3 & 4.

3.0**REFERENCE DOCUMENT :**

- 1) Operation Guidance Note OGN/OPS/ELEC/005.
- 2) 500MW BHEL generator O&M manual

4.0**SUPERSEDED DOCUMENTS:**

LMI/OGN/OPS/ELEC/005.

5.0**SPECIAL INSTRUCTIONS TO BE FOLLOWED:**

Instruction	Responsibility	Document Ref.
Measurement of Dew Point	Chemistry	Work Instruction for Dew Point Measurement
Cold gas temperature Parameter control.	Operation	500MW Generator O&M manual

6.0**METHODOLOGY OF IMPLEMENTATION:**

- Weekly checking of H₂ Gas dew point and reporting in MTP Meeting.
- Maintaining the cold gas temperature and stator water I/L Temperature to cold gas temperature differential by operation.
- During Generator overhaul when rotor is threaded out two NDT Test on Retaining Rings to be carried out
 - (i) Fluorescent Die Penetration Test: This is the most sensitive Method and is capable of detecting very small cracks

(ii) Ultrasonic Test: This method will detect all major defects including cracks down to 3 mm in size. It may, however, be difficult to detect cracks close to surface.

The method is useful because it can be carried out with end rings in situ but should not be relied upon as a guarantee of freedom from all defects.

For further details on DP and ultrasonic testing procedures, advice should be sought from plant FQA department and/or the manufactures.

The material for 500 MW Motor Retaining Ring being XS Cr-Mn NI8-18 K dismantling of such Retaining Ring is not suggested by the Guidance Note.

7.0**FREQUENCY OF CHECKING AND MEASUREMENTS:**

During Generator major overhauling.

8.0**RESPONSIBILITY CENTRE FOR IMPLEMENTATION:**

Weekly Measurement of Dew Point	Chemistry	Chemistry department website Environment reports
Cold Gas Temperature Monitoring	Operation	Operation Logbook
NDT on Retaining Ring during Generator Major O/H	Electrical Maintenance	Generator Test Format

The exception & dispensation of deviations: Exception will be reported to HOD (EM).

9.0**RECORD KEEPING:**

As mentioned above records will be maintained in Chemistry department website/Shift Logbook/Generator Test Format.

10.0**REVIEW:**

Document will be reviewed after Two years by HOP (Simhadri).

11.0**RECORDS OF REVISION:**

- i)Fluorescent Die Penetration test and ultrasonic test described in clause 6.0.
- ii)Checklist added at clause 12.0
- iii)Reference given to Generator O&M manual for cold gas temperature control in clause 5.0
- iv)500MW BHEL generator O&M manual added as reference document in clause 3.0
- v)LMI revised as *OGN/OPS/ELEC/005 revised in June 2020.*

12.0**CHECKLIST:**

Weekly monitoring of H2 dew point temperature.

SL NO	DATE OF SAMPLE	UNIT#1	UNIT#2	UNIT#3	UNIT#4	REMARKS